

Pods, ReplicaSets, and Deployments in Kubernetes

♦ 1. Pod

- A **Pod** is the **smallest and simplest unit** in the Kubernetes object model.
- It represents **a single instance of a running process** in your cluster.
- A Pod can contain **one or more containers** that share:
 - The same **network namespace** (IP, port space)
 - **Storage volumes**

Use case: Think of a Pod as a wrapper around one or more tightly coupled containers (e.g., a main app + a helper sidecar).

♦ 2. ReplicaSet

- A **ReplicaSet (RS)** ensures that a specified **number of Pod replicas** are running at any given time.
- If a Pod dies or is deleted, the ReplicaSet creates a new one to replace it.
- ReplicaSets manage **replication and scaling** of Pods.

Example: You want 3 instances of your web app running. A ReplicaSet will keep 3 Pods alive at all times.

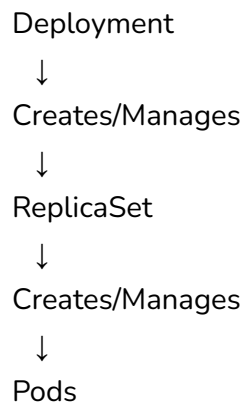
💡 You rarely create ReplicaSets directly. They're usually created and managed by Deployments.

♦ 3. Deployment

- A **Deployment** is a higher-level object that:
 - Manages ReplicaSets
 - Handles **updates**, **rollbacks**, and **rollouts**
- It provides a declarative way to manage Pods and ReplicaSets.

Use case: You define a Deployment that says "run 5 instances of this app, update them gradually when a new version is available, and roll back if something goes wrong."

How They Work Together



Quick Analogy

Think of it like this:

- **Pod** = 1 instance of your app
- **ReplicaSet** = Ensures the correct number of app instances are running

- **Deployment** = Manages the rollout, versioning, and scaling of your app across the cluster
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Sure! Below are example YAML files for:

1. A **Pod**
2. A **ReplicaSet**
3. A **Deployment**

Each uses the same simple app (`nginx`) for clarity.

📄 1. **Pod YAML Example**

```
```yaml
apiVersion: v1
kind: Pod
metadata:
 name: nginx-pod
spec:
 containers:
 - name: nginx
 image: nginx:latest
 ports:
 - containerPort: 80
```
```

📄 2. **ReplicaSet YAML Example**

```
```yaml
apiVersion: apps/v1
kind: ReplicaSet
```

```
metadata:
 name: nginx-replicaset
spec:
 replicas: 3
 selector:
 matchLabels:
 app: nginx
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - name: nginx
 image: nginx:latest
 ports:
 - containerPort: 80
...

```

>  This ReplicaSet keeps 3 Pods running with the `app: nginx` label.

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###  3. \*\*Deployment YAML Example\*\*

```
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:

```

```
    app: nginx
  spec:
    containers:
      - name: nginx
        image: nginx:latest
        ports:
          - containerPort: 80
    ...
```

>  This Deployment will manage the ReplicaSet and handle updates, rollbacks, etc.
